Def. A set $B \subseteq \mathbb{R}^d$ is called a Box if:

$B = I_1 \times \cdots \times I_d$ where $I_i \subseteq \mathbb{R}$ are intervals.

And, $|B| = \ell(I_1) \times \cdots \times \ell(I_d)$ is called a volumn of B.

A set $E \subseteq \mathbb{R}^d$ is called an elementary set if:

$E = \bigcup_{n=1}^{N} B_n$ where $B_1, \cdots, B_N$ are boxes.

Denote $\mathcal{E} := \{E \subseteq \mathbb{R}^d \mid E \text{ is elementary}\}$.

Thm. Fundamental theorem of elementary set.

Given an elementary set $E \in \mathcal{E}$,

E can be expressed as finite union of disjoint boxes.

Proof. Let E be an elementary set of $\mathbb{R}^d$. That is,

$E = \bigcup_{n=1}^{N} B_n$ for some boxes $B_1, \cdots, B_N$.

For each $1 \le i \le N$, denote that $B_i = I_{i,1} \times \cdots \times I_{i,d}$.

